Complex Homework #10

1. Let $U \subset \mathbb{C}$ be open and connected. Let $f_n \to f$ be holomorphic functions, where the convergence is uniform on compact subsets. Suppose f is not identically zero. Prove that if $f(w) = 0$, then we can write $w = \lim z_n$, where $f_n(z_n) = 0$ for all n sufficiently large. Does this hold if we assume only that U is open (but not that it is connected)?

Since we have uniform convergence on compact subsets we have that f is holomorphic[1] on U. By isolation of zeroes there's some neighbourhood of w contained in U where f is nonvanishing. Now consider a slightly smaller neighbourhood about w whose closure is contained in the ball of nonvanishing f. Call this final ball B. In particular on ∂B we have $|f(z)| \geq \delta$ for some positive δ. On the other hand, for sufficiently large n we have that $|f_n(z) - f(z)| < \delta$ for all $z \in \partial B$ since the convergence is uniform, and so $|f(z)| > |f_z(z) - f(z)|$ on ∂B. We have met the conditions for Morera's theorem and so it follows that "$f + g$" $= f_n(z)$ has the same number of zeroes, counted with multiplicity, in B as $f(z)$.

By construction, we have that f has exactly one zero in B. And so for each n sufficiently large we have that f_n has exactly one zero contained in B. Call this sequence $\tilde{z}_n$. Since $z_n \in \overline{B}$ a compact set, we have that some subsequence of $\tilde{z}_n$ converges to some point in $z_* \in \overline{B}$, call this subsequence z_n. Now, since $f_n \to f$ and f has exactly one zero at w we must have that $z_n \to w$. Indeed

$$0 = \lim_{n\to\infty} f_n(z_n) = f(z_*),$$

but $f = 0$ only at $w \in \overline{B}$, and so $z_* = w$. The z_n is our desired sequence.

The argument above only used constructions that were local about the point w where f vanished, and so I think this conclusion still holds if U is a non-connected open set, given that f is not identically zero on any component. The existence of the nonvanishing neighbourhood about w depends on the fact that f is nonvanishing. Given the wording

[1] Consider applying Morera's theorem to small rectangles about each point, and since the convergence is uniform on compact subsets we can exchange the limit and the integral.

of the hypothesis, we could have an f which is nonvanishing on all of U but vanishes identically on some connected component. On that component this argument would not work.

$\blacksquare$

2. Determine the automorphism group of $\mathbb{C} - \{0, 1\}$.

Consider for a moment holomorphic functions on the set $X := \mathbb{C} \setminus \{0, 1\}$. We can consider the holomorphic functions on X as meromorphic functions on $\mathbb{C}$ with potentially singularities at $0, 1$. Notice that any $f \in Aut(X)$ cannot have an essential singularity at either $0, 1$, otherwise Castorati-Weirstrass implies that f would not be injective (its image would hit almost every value of $\mathbb{C}$ infinitely often). That is, any $f \in Aut(X)$ "at worst" has a pole at $0, 1$. It could also extend to a holomorphic function at either of these points. In other words, a priori $f : X \to X$ is bijective and holomorphic, however, we can instead think about it as a function $f : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$. Moreover, by definition, the inverse of any $f \in Aut(X)$ is also holomorphic on X. And so by the same argument we can also think of f^{-1} as a function $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$.

First consider the case where f extends to a holomorphic function at either $0, 1$. Since f still needs to be bijective on X, if f is extends to a holomorphic function at $0, 1$, then it can only either fix these points or send them to each other. More precisely, suppose f extends to a holomorphic function at both $0, 1$. Then we must either have that f fixes both $0, 1$ or f exchanges $0, 1$. The first case is satisfied by only the identity $f(z) = z$ and the second is satisfied by $f(z) = -z + 1$. The latter was found by recalling that in such a case f is a bijective holomphism of the entire plane and we found these to be of the form $f(z) = az + b$.

Now, suppose that f does not extend to a holomorphic function at both $0, 1$. Then we argued above that we can think of f, f^{-1} as being extended to holomorphic maps $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$. Notice that the extended f, f^{-1} must still be inverses of each other as functions $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$, since they agree with their original versions on the dense open set $\hat{\mathbb{C}} \setminus \{0, 1, \infty\}$. In other words, we can moreover view f, f^{-1} as bijections of $\hat{C}$. In other words, we can view f, f^{-1} as Mobius transformations.

A pole at either $0, 1$ corresponds to f sending either $0, 1$ to $\infty \in \hat{\mathbb{C}}$. But since we argued that f, f^{-1} must be bijections on $\hat{\mathbb{C}}$, it must be that f only has a pole at exactly one of

$0, 1$. That is, we can view f as a Mobius transformation which sends the triple $(0, 1, \infty)$ to some permutation of the same triple. It must send this triple to some permutation of itself, because we need f, f^{-1} to be functions on $X = \mathbb{C} \setminus \{0, 1, \infty\}$. Recall that the Mobius transformations are triply transitive, and so, once we specify which permutation the triple $(0, 1, \infty)$ is sent to, then we have exactly determined the Mobius transformation.

We have argued that we have an automorphism of X for each permutation of the triple $(0, 1, \infty)$ (i.e. there are 6 such automorphisms).

We can compute the form of these maps exactly. Consider the automorphism of X which sends $(0, 1, \infty)$ to $(\infty, 1, 0)$. We start with a Mobius transformation $f(z) = (az + b)/(cz + d)$. The condition $f(0) = \infty$ implies $d = 0$. The condition $f(1) = 1$ implies $c = a + b$. Then lastly the condition $f(\infty) = 0$ implies $a = 0$ and that one automorphism of X is $f(z) = 1/z$. Proceeding in a similar way for the other permutations gives us the following list for the automorphisms of X. The right column indicates where the triple $(0, 1, \infty)$ is sent to.

$$
\begin{aligned}
f(z) &= z && (0, 1, \infty) \\
f(z) &= \frac{z}{z-1} && (0, \infty, 1) \\
f(z) &= 1 - z && (1, 0, \infty) \\
f(z) &= \frac{1}{1-z} && (1, \infty, 0) \\
f(z) &= \frac{z-1}{z} && (\infty, 0, 1) \\
f(z) &= \frac{1}{z} && (\infty, 1, 0).
\end{aligned}
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3. Show that

$$z \mapsto \int_0^z \zeta^{-\beta_1}(1-\zeta)^{-\beta_2} d\zeta,$$

with $0 < \beta_1, \beta_2 < 1$ and $1 < \beta_1 + \beta_2 < 2$, maps $\mathbb{R} \cup \infty$ to a triangle. What happens when $\beta_1 + \beta_2$ is 1, and what happens when it is < 1?

To set the stage, recall that the image of this is piecewise linear with "turns" at $z = 0$ and $z = 1$ where the angle of turning is given by $\beta_1 \pi$ and $\beta_2 \pi$.

Let us first discuss the boudnary cases. In general, when $|z|$ is large we have that this integral goes like

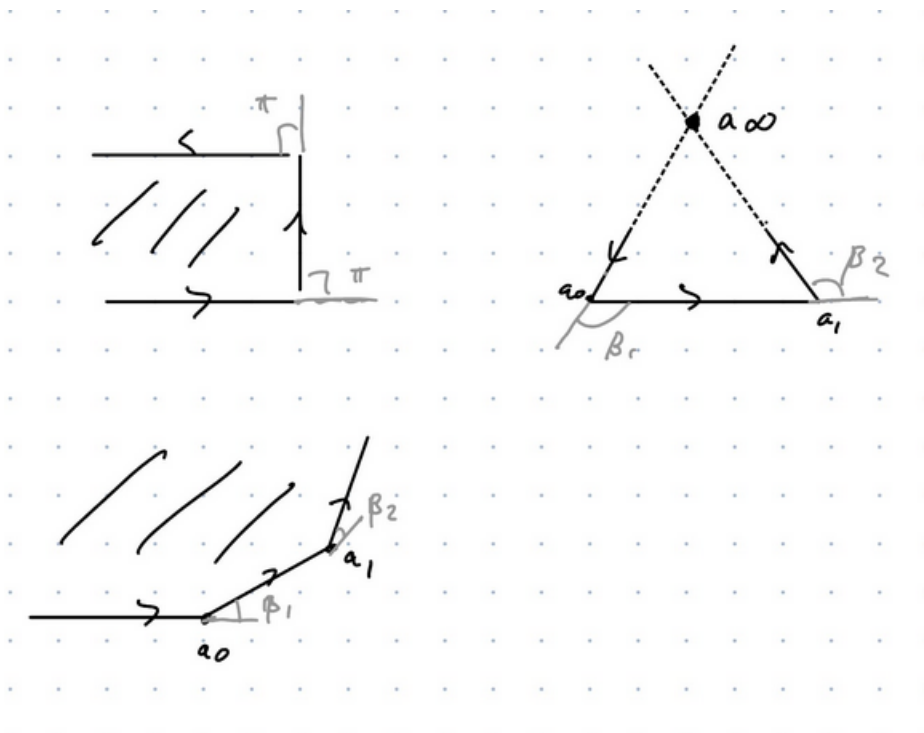
$$f(z) \sim \int \zeta^{-(\beta_1+\beta_2)} d\zeta.$$

In the case where $\beta_1 + \beta_2 \leq 1$ we have that this integral does not converge and so "the image of ∞ " is not finite. In other words we get the the image of $\partial\mathbb{H}$ is a piecewise union of straight lines, not bounded. In particular, if $\beta_1 + \beta_2 = 1$ then we have that the angles are both right angles and we recover the case of the unbounded strip that we saw in class. On the other hand, if $\beta_1 + \beta_2 < 1$ then we have that $\beta_1, \beta_2 < 1$ and then we have an "obtuse" unbounded strip.

Now if $\beta_1 + \beta_2 > 1$ we have that $f(\infty)$ is some finite point. Recall also that the integrals given by $f(\infty)$ and $f(-\infty)$ converge to the same point (we showed this in class by considering a contour integral of an expanding semicircle in the upper half plane). Consider $\beta_1 + \beta_2 = 2$, in particular $\beta_1 = 1$ and $\beta_2 = 1$. In this case the image is a line segment — for z between $-\infty$ to 0 we have a straight line segment, then a turn of π and then a straight line segment for z from 0 to 1, a turn of π again, and then a straight line segment.

Now consider what happens when $1 < \beta_1 + \beta_2 < 2$ we know that the image of $(-\infty, 0)$ is a straight line segment, followed by a turn of $\beta_1 \pi$, followed by another straight line segment for $(0, 1)$, a turn of $\beta_2 \pi$, and then another line segment for $(1, \infty)$. The fact that these line segments connect is given by the fact that the definition of f gives a continuous function. We know from prior argument that $f(\infty) = f(-\infty)$ and so the image of the entire boundary $\mathbb{R} \cup \infty$ is a closed polygon with three vertices (i.e. a triangle).

See the figures below.



Here are some images of different cases. The top left image shows when $\beta_1 + \beta_2 = 1$ and the lower left image shows $\beta_1 + \beta_2 < 1$. The images here are noncompact. The image on the right shows the situation when $1 < \beta_1 + \beta_2 < 2$. In principle $a_{-\infty}$ and a_∞ must lie on the dotted line segments, since the derivative of the given map is smooth away from $0, 1$. However, since $a_\infty = a_{-\infty}$ we have that the value at infinity must be the intersection of these two dotted lines, giving the image of the boundary is indeed a triangle.

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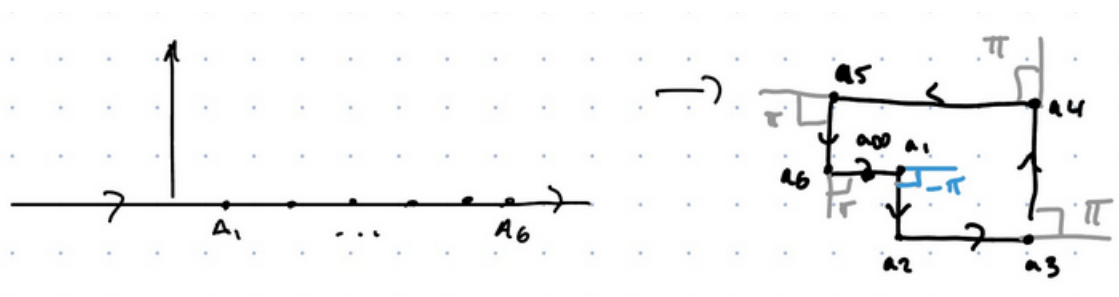
4. Find an example of a holomorphic map $F : \mathbb{H} \to \mathbb{C}$ that extends to a continuous map on $\overline{\mathbb{H}} = \mathbb{H} \cup \mathbb{R} \cup \{\infty\}$ that maps $\mathbb{R} \cup \infty$ bijectively to the boundary of a *non-convex* polygon.

We have seen that for some points $A_1, \cdots, A_n \in \mathbb{R}$, and some $\beta_1, \cdots, \beta_n \in \mathbb{R}$ gives a Schwarz-Christoffel map

$$f(z) = \int_0^z (\zeta - A_1)^{-\beta_1}(\zeta - A_2)^{-\beta_2} \cdots (\zeta - A_n)^{-\beta_n} d\zeta.$$

whose image of $\mathbb{R} \cup \infty$ a series of line segments, which forms a polygon under appropriate assumptions for the β_i (see previous question for example). The polygon given has exterior angles $\pi\beta_i$.

Consider the choice of $\beta_1 = -1$, and then $\beta_i = 1$ for $n = 2, \cdots, 6$. Choose any $A_i \in \mathbb{R}_+$, say. Thinking about varying z continuously from 0 to ∞, the image of this map has one "turn to the right" by π, and then the rest of the turns "are to the left" by π (see the figure below). Here the choice of the $A_i \in \mathbb{R}$ does not matter, since I think there is no way for this polygon to be self-intersecting with this particular choice of angles.



This image shows the map and its output. Here, the turning angle is π except at A_1 which has turning angle $-\pi$. There is no way to get this polygon to self intersect with these angle constraints.

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5. Find a biholomorphism $f : \Delta \to \mathbb{C} - [1, \infty)$ with $f(0) = 0$ and $f'(0) > 0$.

First recall that the upper half plane is biholomorphic to the disc via $z \mapsto i \cdot (1+z)/(1-z)$. Next the map $z \mapsto z^2$ "almost gets us what we want" since its image on the upper half plane is $\mathbb{C} \setminus (0, \infty)$. Note that this is indeed a biholomorism from the upper half plane to $\mathbb{C} \setminus (0, \infty)$ since the target does not contain 0, and we can take the branch cut along $(0, \infty)$ to define our square root function as an inverse. Lastly, we can apply a translation $z \mapsto z + 1$ so that the output is finally $X = \mathbb{C} \setminus (1, \infty)$.

To summarize, the proposed map is

$$(z \mapsto z+1) \circ (z \mapsto z^2) \circ \left(z \mapsto i \cdot \frac{1+z}{1-z}\right) = z \mapsto 1 - \frac{(1+z)^2}{(1-z)^2}.$$

Plugging in $z = 0$ into the above map shows that its image is zero: $0 \mapsto i \mapsto -1 \mapsto 0$. If we call our proposed function $\tilde{f}$, then an application of the quotient rule and a couple applications of the chain rule will give us that

$$\tilde{f}'(z) = \frac{-4(1+z)}{(1-z)^3},$$

i.e $\tilde{f}'(0) = -4 < 0$. And so, if we want $f'(0) > 0$ then we need to ammend this map slightly. Note that by the chain rule, $\frac{d}{dz}\tilde{f}(-z) = -\tilde{f}'(z)$. And so if we can precompose $\tilde{f}$ by negation then we will have fixed our derivative condition. And indeed, recall that rotations are biholomorphisms of the unit disc. And so $f(z) := \tilde{f}(e^{\pi i} z) = \tilde{f}(-z)$ is still a biholomorphism from Δ to X. And so overall

$$f(z) = 1 - \frac{(1-z)^2}{(1+z)^2}$$

is a biholomorphism from $\Delta \to X$ which sends 0 to 0 and for which $f'(0) = 4 > 0$.

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6. Let $\mathcal{S} \subset \mathbb{C}$ be the interior of a square. Find an explicit example of a diffeomorphism $F : \Delta \to \mathcal{S}$ that does not have a continuous extension $\overline{\Delta} \to \overline{\mathcal{S}}$.

Allow me to connect some ideas here briefly. We saw in class that any biholomorphism from $\Delta \to \mathcal{P}$ for some polygon $\mathcal{P}$ extends continuously to a homeomorphism $\bar{\Delta} \to \bar{\mathcal{P}}$. So here, we are allowing a more general class of functions, bijective smooth functions $\Delta \subseteq \mathbb{R} \to \mathcal{P}$ which behaves badly enough on the boundary to not be continuously extendable. In particular, whatever function we come up with should not be holomorphic.

To begin, consider a smooth bijection $\Delta \to \Delta$ where a circle of radius $r < 1$ is mapped to the same circle, but rotated by some $\theta(r)$ where $\theta \to \infty$ as $r \to \infty$. For example, consider $f : \Delta \to \Delta$ via

$$f(z) := e^{\frac{i}{1-|z|}} z \qquad \text{or} \qquad f(r, \theta) := (r, \theta + 1/(1-r)).$$

This function is infinitely differentiable when viewed as a function from $\mathbb{R}^2 \to \mathbb{R}^2$. Viewed from the polar coordinate lens, all the partials exist and are continuous, away from $r = 0$, since the transformation is the identity in r and a translation in θ. For $r = 0$ we could rewrite the function in cartesian coordinates, (the definition would now involve some $\sin, \cos$ terms), and we would find that the function is still smooth. Moreover, this map is bijective because it has an inverse given by

$$f^{-1}(z) = e^{\frac{-i}{1-|z|}} z.$$

On the other hand, this function does not have a continuous extension to the boundary of Δ, because the rotations diverge as we approach the boundary.

We have a diffeomorphism $\Delta \to \Delta$ which does not extend continuously to the boundary. Now to find the desired function we can compose f with our biholomorphism $\Delta \to \mathbb{H}$ via $z \mapsto i \cdot (1+z)/(1-z)$, and then compose this with the Schwarz-Christoffel map which sends $\mathbb{H} \to \mathcal{P}$. Recall that holomorphic function is always also smooth when viewed as a real vector function, and so the composition would give a diffeomorphism $\Delta \to \mathcal{P}$ which does not extend continuously to $\bar{\Delta}$.

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